

No. of Printed Pages : 4
Roll No.

220965A

6th Sem/ Electrical
Subject : HVDC & Flexible AC Transmission Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 One major advantage of HVDC over HVAC is: (CO1)

- a) No reactive power flow
- b) High charging current
- c) High skin effect
- d) Complex protection

Q.2 Conversion of DC to AC in HVDC system is done by: (CO2)

- | | |
|--------------|----------------|
| a) Rectifier | b) Transformer |
| c) Inverter | d) Reactor |

Q.3 Which component is used to smooth DC current in HVDC? (CO2)

- | | |
|--------------------|----------------------|
| a) Filter | b) Smoothing reactor |
| c) Circuit breaker | d) Capacitor bank |

- Q.4 SVC is mainly used for : (CO4)
- a) Series compensation
 - b) Voltage & reactive power control
 - c) Power generation
 - d) Frequency control
- Q.5 FACT controllers improve power system: (CO3)
- a) Cost
 - b) Stability
 - c) Weight
 - d) Insulation
- Q.6 Which FACTS controller can control power flow and voltage simultaneously? (CO4)
- a) SVC
 - b) TCSC
 - c) STATCOM
 - d) UPFC

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 DC link is used for _____ transmission of power. (CO2)
- Q.8 FACTS stand for _____. (CO3)
- Q.9 Series compensation reduces line _____. (CO4)
- Q.10 HVDC transmission is economical for long distance power transfer. (True/False) (CO1)
- Q.11 TCSC is a shunt FACTS controller. (True/False) (CO4)

Q.12 Filters are required in HVDC systems to suppress harmonics. (True / False) (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain need of HVDC transmission. (CO1)

Q.14 Write applications of DC transmission system. (CO1)

Q.15 Explain principle of DC link control. (CO2)

Q.16 Explain role of smoothing reactor in HVDC. (CO2)

Q.17 Define FACTS controllers and list their types. (CO3)

Q.18 Explain objectives of series compensation. (CO4)

Q.19 Explain working of Static VAR compensator (SVC). (CO4)

Q.20 Write short note on TCSC. (CO4)

Q.21 Explain principle of UPFC. (CO4)

Q.22 What is meant by a Back-to-Back HVDC system? (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain HVDC transmission system and its advantage over HVAC. (CO1)

- Q.24 Explain FACTS technology and its role in power system performance. (CO3)
- Q.25 Explain the Shunt and Series Compensation with diagrams. (CO4)